

Spark daemon pattern: review package (r4.8)

- **For:** reviewers (people or models). This document stands alone. Every file it names is in spark-daemon-pattern/ on branch claude/clever-bardeen-qgkxez, for reviewers who can read the repository.
- **Status:** FOR REVIEW. It authorizes no activation, no class unlock and no LTC change. Verdicts marked "Claude, subject to the owner" stand unless the owner overturns them. Items marked **OWNER** need the owner's own word.
- **Revision:** r4.8, 2026-10-06, by Claude.
- **State at r4.8:**
 - contract 3.1.0 (unchanged since r4.5);
 - 214 of 214 self-tests pass, none skipped (Python 3.12, with jsonschema 4.26 for the four schema-agreement tests; evidence/r4.8/selftests.txt);
 - the four reference daemons pass all 19 battery checks (DB-01 to DB-18 and DB-20) on the build workspace, Landlock ABI 7 (evidence/r4.8/battery-examples.txt);
 - the pattern has been ported to its **first pilot project** and merged there, with one observer daemon (\$2).
- **Supersedes:** the status and seam tables of REVIEW-PACKAGE-COMBINED.md (r4.4) and REVIEW-PACKAGE-R4.5.md (r4.5). The detail in those documents still holds unless this one says otherwise; it is cited, not repeated.

How to mark: for each decision, seam, gap or question, give AGREE, CHANGE or REJECT, with notes. Say which kind of evidence you rely on: **reproduced**, **measured**, **traced** (to code) or **opinion**. §8 lists the questions this review most needs answered.

Reading order:

1. §1 and §2: what changed since r4.5, and what the first deployment teaches.
2. §3 to §5: where every decision stands, what is owed to the owner, and the seams. **This is the main request.**
3. §6 and §7: the battery and the self-test list.
4. For detail: HARDENING-REVIEW-R4.6.md, ADJUDICATION-R4.5.md, BATTERY-REVIEW-R4.5.md, SQLITE-LEDGER-REVIEW.md, BREAK-GLASS.md, LTC-INTEGRATION-REVIEW.md, PD-01-DAEMON-CLASSES.md, HARDENING.md.

1. What changed since r4.5

Revision	Change	Kind	Evidence
r4.6	HF-36 (Medium). The Landlock domain granted execute wherever it granted read, including the output directory. A binary written there by a compromised daemon, or dropped by someone else into a watched folder, could be run if the in-process layers were bypassed. Execute now stays on system paths only	Defect, fixed	Reproduced under ABI 7 (evidence/r4.6/exec_before.txt); tests/test_landlock.py::test_nothing_the_daemon_can_write_or_merely_reads_can_be_executed
r4.6	PD-93 to PD-98 adjudicated: no disassembly gate (digest-pinned executables instead); an activation-time ELF provenance check; containment of untrusted text; credential stores denied; the supervisor alone holds the ledger; F-1 closed by a check after opening	Decisions (Claude, subject to the owner)	HARDENING-REVIEW-R4.6.md §4
r4.6	F-1 recorded. A directory swapped for a symlink between ctx.read_text's path check and its open can reach a denied subtree inside a declared read. systemd's InaccessiblePaths= still blocks it under the	Weakness, fix planned	Simulated deterministically (evidence/r4.6/read_race_simulation.txt)

Revision	Change	Kind	Evidence
	unit; outside the unit it does not		
r4.7	HF-37 (Medium). Precheck bound the original manifest instead of the workspace copy. A daemon with an absolute output_dir failed precheck ("no ledger") while the battery passed it 19 of 19. Found while porting the pattern to the pilot	Defect, fixed	Reproduced by the pilot's own daemon; tests/test_handoff.py::test_an_absolute_output_dir_prechecks_like_the_battery
r4.8	Two unit options from the pilot. unit --require-path adds ConditionPathExists=, so a daemon writing to a locked drive is skipped rather than stopped fail-closed. unit --part-of adds PartOf=, After= and WantedBy= another unit, so a daemon whose open ledger would keep a drive busy stops before the unit that owns the drive and starts with it. The unit name is checked, so it cannot inject a directive	Feature, from deployment	tests/test_misc.py::test_a_required_path_becomes_a_start_condition, test_part_of_binds_the_unit_to_another
r4.8	proc.SYSTEM_PATH replaces the command search path written out in 17 places; tests/conftest.py; git-watch's Git calls are counted by the parser, not by text	Hygiene	214 of 214 self-tests

2. The first deployment, and what it teaches

The pattern was copied into one pilot project, not linked to it. The pilot runs one class-1a observer, pressure-watch. It records when a single host's available memory, swap in use or drive free space crosses that project's own soak limits, and nothing else. All figures below come from that copy. The pilot's name and paths are deliberately left out of this repository.

What the pilot's owner decided (their decisions D-1 to D-4, 2026-10-01):

#	Question	Decision	What it means for the pattern
D-1	The generated unit restarts on a crash; that project's own units never restart	Keep the restart , and tell the owner about every restart and every stop that needs a person, in the project's chat	The restart policy (RestartPreventExitStatus=2 65 73 78, 5 starts in 300 s) survived contact with a stricter house rule, on condition of a notifier . That is Phase 1's OnFailure= notifier, built by the pilot, not here (seams N-19, N-22)
D-2	Where the unit lives	Generated on the host at install time; committed only after a soak	The generator, not a checked-in unit, is the source of truth
D-3	The observer overlaps an existing 5-minute sampler	Keep both during the pilot	—
D-4	strace on the host	Installed , so DB-08 and DB-09 can run there	A host battery can now reach PASS rather than INCOMPLETE (G-7)

How the pilot activates it. The project's installer runs on setup day and on every update. For a fixed list of observers (today only pressure-watch) it:

1. runs the full battery **on the host, as the service user**;
2. on PASS only, generates the unit with --require-path (the drive's marker) and --part-of (the project's stack unit), installs and enables it;
3. on anything else, installs nothing, disables a unit installed before, and warns with the report's path.

The owner's Class C decision has therefore been **delegated to a battery PASS on the host**, rerun at each update. This is a design choice the pilot's owner made with the facts in front of them. It also makes three of this repository's open items load-bearing sooner

than planned: PD-40 (the register), PD-93 (digest-pinned executables) and G-5 (a PASS bound to its host) (seam N-20).

What is still not known: the first host battery and the first DAEMON_START under systemd as PID 1 have **not been reported back** to this repository. G-6 ("never run under systemd PID 1") stays open until that DAEMON_START.landlock.status is read and is "enforced".

3. Where every decision stands

Status labels: **Built** (in code, with tests); **Partial**; **Designed** (documents only); **Planned** (adjudicated, scheduled, not started). "Owner" means the decision needs the owner's word before it is final.

3.1 Built

Item	Where	Since
PD-63 blind period, SENSE_BLIND exit 78, never restarted	runtime.py, context.py, unitgen.py	r3.5
PD-70 blindness across restarts, on CLOCK_BOOTTIME within a boot (HF-34)	runtime.py (_inherited_blindness, _boot_stamp)	r4.0, r4.5
DB-20 the daemon observes (HF-35)	battery.py (db20)	r4.5
HF-33 capped listing fails the cycle (dir-watch only)	examples/dir-watch/daemon.py	r4.3
HF-36 execute only on system paths	landlock.py	r4.6
HF-37 precheck binds the workspace manifest	handoff.py	r4.7
PD-95 containment of untrusted text in the digest	render.py, DB-10	r3
--require-path, --part-of	unitgen.py, cli.py	r4.8
Contract 3.1.0	contract.py, contract/	r4.5

3.2 Planned, in the order adjudicated

#	Item	Lands in	Notes
1	F-1 check after opening, plus 0_NOFOLLOW on the ledger, lock and digest opens (PD-98)	context.py, guard.py, ledger.py, runtime.py	Next. The only weakness shown and not yet closed
2	PD-93 each allowlisted executable's SHA-256 in DAEMON_START.tools	runtime.py	Additive. Becomes urgent where activation is automated (\$2, N-20)
3	Phase 1: an OnFailure= notifier; a change-detecting staleness checker (N-5); a budget report	new	The pilot built its own notifier (D-1); compare before building (Q-3)
4	DB-19 direct cgroup memory peak, with a separate helper, and the CI rule for INCOMPLETE	battery.py + helper	OWNER wording (PD-76). Replaces DB-14's memory half (38 % under-read, measured r4.4)
5	Contract 4.0.0 in one batch: exit split (PD-63), battery codes 0/3/4 (C-1), truncation always raises (PD-83), pacing with jitter as default (PD-85), consequence field (PD-01.2), credential stores and secret-shaped names denied (PD-96), PD-35/39/49, schema-3 migration hint	many	One migration for every author (N-13)
6	DB-21 budget table (PD-69, PD-86); DB-22 independent ledger verifier (B-6); DB-23 worst-case fixtures (PD-71); PD-94 executable provenance check	battery.py, a verifier outside the package	—
7	PD-40 activation register; PD-72 supervisor/worker split (with PD-97); PD-01.7 connector gate	new	Preconditions for every class above 1a
8	Break Glass beacon tier (PD-01.10 to PD-01.14, BG-S)	new	After PD-01.3, PD-72 and PD-01.7. The acting tier (BG-A, class 3b) is deferred with kernel v2

3.3 Designed, or adopted as rules, with nothing to build yet

- **Package A (transport):** PD-82 bind, do not embed; PD-84 register vocabulary; PD-86 retry bounds over time; LI-1 to LI-14; the LTC-H01 to H03 markup, forwarded to the LTC reviewers.
- **SQLite ledgers:** PD-87 observers never govern; PD-88 one set of hash-chain rules; PD-89 wake is a hint; PD-90 UDS profile guidance; PD-92 the three-ledger split by writer. **Deferred:** PD-91, until the kernel's ledger exists and the read-only WAL problem is solved on the target's SQLite.
- **Classes:** PD-01.1 to PD-01.3; BF-1 to BF-6 (BF-5 mandatory); A-1 to A-12 (A-8 to A-10 deferred).
- **Deferred:** WASI with fuel (PD-73, PD-74); where T is set (PD-63, until PD-40).

4. What is owed to the owner

Item	What is needed	Blocks
PD-01.9 scope (OWNER)	Confirm: the Blind Forester does not unlock critical; BF-5 is mandatory where a person could be harmed; it governs survival mode, never emergency authority; PD-72 is a prerequisite	Every active class

Item	What is needed	Blocks
PD-76 wording (OWNER)	Confirm "a direct cgroup peak: v2 memory.peak or v1 memory.max_usage_in_bytes", with a v1 reading admissible only against a v1 limit	DB-19
Delegated verdicts	Ratify or overturn: r4.5 (ADJUDICATION-R4.5.md: settled-ruling amendments, Packages A to C, PD-87 to PD-92) and r4.6 (PD-93 to PD-98)	Nothing today; each stands until overturned
The host report	The first host battery report and the first DAEMON_START from the pilot	G-5, G-6

5. Seams

A seam is a boundary where two parts meet and each assumes something about the other. Every defect found since r3.5 sat on one. Test each with: "What does each side assume about the other, and what happens when that assumption is false?"

N-1 to N-14 (REVIEW-PACKAGE-COMBINED.md, reviewer focus) and **N-15 to N-18** (SQLITE-LEDGER-REVIEW.md §5) stand as adopted, each becoming a required test when its decision is built. N-10 was removed by PD-83's modification.

New at r4.8, from the first deployment. Each was found by reading the pilot's code (**traced**). None has been reproduced.

#	Seam	The assumption that could fail	What a reviewer should check
N-19	A ledger consumer outside the pattern (the pilot's notifier)	The notifier reads each daemon's ledger to report restarts. It reads a payload flag (previous_run_ended_cleanly) without verifying the hash chain. It reports stops from systemd's unit state, which is independent	Can a malformed or edited record make a false "restarted" notice, or hide a real one? Should every consumer either verify the chain (DB-22's independent verifier, once built, is the natural library) or treat what it reads as a hint only? The notice text is fixed words and checked fields, never the daemon's own text: confirm that holds for every field
N-20	Automated activation (the pilot's installer; PD-40, PD-93, G-5)	The code the battery judged is the code the unit runs	DAEMON_START records manifest_sha256 and daemon_code_sha256, but nothing compares them with the PASS report's digests. A repository change that reaches the host by any path other than the installer runs at the next restart with no battery. Should the runtime, or a pre-start step, refuse a start whose digests differ from the last PASS on this host? This is PD-40's job; is a minimal check owed before PD-40?
N-21	PartOf= and the meaning of a heartbeat (r4.8)	"Heartbeats prove it was watching while nothing changed"	When the owning unit stops, the daemon stops cleanly, so no blindness is inherited at the next start (correct under PD-70). The ledger shows the gap as DAEMON_STOP to DAEMON_START. Does every reader of "coverage" treat that gap as not watching , rather than as quiet?
N-22	ConditionPathExists= and silence (r4.8)	A skipped unit is harmless	A condition failure is not a unit failure: systemd records it, but nothing is written to the ledger and the pilot's notifier, which looks for failed units, says nothing. A drive left locked means the observer silently never starts. Does the staleness checker (Phase 1, N-5) cover "never started", and not only "stopped changing"?
N-23	Two copies of one pattern (no link allowed)	A fix in one copy reaches the other	r4.8 brought the pilot's changes back by hand. HF-36 and HF-37 went the other way by hand. Is a per-revision parity list (change, in source?, in pilot?) enough, or should each revision's tests run against both copies?

6. The conformance battery at r4.8

The checks are unchanged since r4.5: **DB-01 to DB-18 and DB-20**; DB-19 is reserved for PD-76. Each check's method, its limits ("cannot catch") and whether it is independent of the code it judges are in REVIEW-PACKAGE-R4.5.md Part 1 §2, which still holds. Two changes in what is known:

- **DB-08 and DB-09** can now run on the pilot host (D-4).
- **The battery is now an activation gate** in the pilot (§2). That raises the cost of the known gaps:

#	Gap (from r4.5)	Change at r4.8
G-1	Memory read per process (DB-14), 38 % low	Unchanged. DB-19 waits on the PD-76 wording (OWNER)
G-2	Ledger checks use the skeleton's own verifier	Unchanged; N-19 adds a second reason for DB-22
G-3	Limits not checked against each other	Unchanged (DB-21)
G-4	Fixtures are small	Unchanged (DB-23)
G-5	A PASS is host-specific but does not say so	Raised. In the pilot the PASS is produced on the host it gates, which helps, but the report still does not bind a host fingerprint

#	Gap (from r4.5)	Change at r4.8
G-6	Never run under systemd PID 1	Pending the pilot's host report (§4)
G-7	Two checks need tools	Closed on the pilot host (D-4); open elsewhere
G-8	Power-loss durability not provable by SIGKILL	Unchanged

Planned checks are as listed in REVIEW-PACKAGE-R4.5.md Part 1 §4 (DB-19, DB-21, DB-22, DB-23, DB-2x), plus PD-94's executable provenance check. **Planned self-tests** are as listed in its §5 (T-EX1, T-EX2, T-TR1, T-PC1, T-ST1, T-CG1, T-N1 to T-N14), plus one per new seam: T-N19 to T-N23.

7. The self-test list

214 tests at r4.8, generated from the test files by `evidence/r4.5/testlist.py` into `evidence/r4.8/testlist.md`, which is appended below. "Cites" lists the defect (HF), decision (PD), battery check (DB) and rule identifiers each test names. New since r4.5: the HF-36 and HF-37 tests (r4.6, r4.7) and the two unit-option tests (r4.8).

All 214 ran on this revision; none was skipped (`evidence/r4.8/selftests.txt`). Without `jsonschema` installed, the four schema-agreement tests skip by name and say so.

8. Questions for the reviewer

- N-20:** with activation automated in the pilot, should a minimal "digests match the last PASS on this host" check come **before** PD-40, or is it acceptable to wait for the register?
- N-19:** should the pattern publish a small, independent ledger reader (DB-22's verifier) for consumers, and require consumers to use it?
- D-1 and Phase 1:** the pilot built its own restart and stop notifier. Should Phase 1's `onFailure=` notifier be built here as the reference, or should the pattern only specify what a notifier must report (stops by reason, restarts by unclean start, condition skips)?
- N-22:** is "never started" a staleness case the Phase 1 checker must detect, and against what clock?
- F-1:** the adopted fix opens the file, asks the kernel which file the descriptor actually names, and re-checks that path against the policy before reading a byte. Is that sufficient, or should reads go through `openat2` with `RESOLVE_NO_SYMLINKS` where the kernel supports it, with the post-open check as a fallback? (The audit hook judges relative paths, which is why a per-component `O_NOFOLLOW` walk was not chosen: HARDENING-REVIEW-R4.6.md §3.8.)
- G-5:** which host facts must a PASS bind (kernel, Landlock ABI, cgroup version, systemd version, tool versions) before an installer may act on it?
- Which of the 214 self-tests would you delete as redundant, and which property has no test at all?

Markup table

#	Item	AGREE / CHANGE / REJECT	Evidence type	Notes
1	§1 changes since r4.5			
2	§2 lessons from the first deployment			
3	§3.2 order of planned work			
4	§4 owner items			
5	N-19			
6	N-20			
7	N-21			
8	N-22			
9	N-23			
10	§6 gap changes			
11	Q-1 to Q-7			

Appendix: the self-test list

`tests/test_blind.py` (29 tests)

Blind period (r3.5): `ctx.unsettled()`, `blind_limit_seconds` and `SENSE_BLIND`, and the unit

Class	Test	What it pins	Cites
BlindPeriodTests	test_unsettled_is_neither_an_event_nor_an_error	Unsettled is neither an event nor an error.	
BlindPeriodTests	test_always_unsettled_stops_at_the_limit	Always unsettled stops at the limit.	
BlindPeriodTests	test_failed_cycles_count_towards_the_limit	r2's silent-outage shape: a daemon failing every cycle kept pinging the watchdog forever.	
BlindPeriodTests	test_an_accepted_cycle_resets_the_clock	Busy for about twice the limit in total, but never for a whole limit at once.	
BlindPeriodTests	test_blindness_survives_a_restart_loop	HF-32: killed and restarted every 1.0 s against a 1.5 s limit. On r3.9 no run ever reached the limit; now the second run inherits the first run's blindness.	HF-32
BlindPeriodTests	test_a_backward_clock_step_does_not_hide_blindness	On r4.4 every restart after a one-hour backward step inherited 0 ms, and six kill-restarts (6 s blind against a 1.5 s limit) never reached SENSE_BLIND (evidence/r4.5/).	HF-34
BlindPeriodTests	test_start_heartbeat_and_stop_carry_the_boot_stamp	Start heartbeat and stop carry the boot stamp.	HF-34
BlindPeriodTests	test_across_a_reboot_the_wall_clock_is_used_and_a_backward_one_assumes_the_worst	Across a reboot the wall clock is used and a backward one assumes the worst.	HF-34
BlindPeriodTests	test_an_event_after_the_last_stamp_is_anchored_at_that_stamp	A daemon event carries no boot time; the preceding stamp bounds it from below (safe side).	HF-34
BlindPeriodTests	test_a_clean_stop_does_not_reset_blindness	A clean stop does not reset blindness.	HF-32
BlindPeriodTests	test_a_restart_past_the_limit_gets_one_reacquisition_cycle	A restart past the limit gets one reacquisition cycle.	HF-32
BlindPeriodTests	test_heartbeats_tell_quiet_from_dead	A healthy daemon that sees no change still leaves evidence every limit/2.	HF-32
BlindPeriodTests	test_a_malformed_reason_is_a_sense_failure	A malformed reason is a sense failure.	
BlindPeriodTests	test_the_override_is_ignored_outside_test_mode	Outside test mode the manifest's 600 s applies, and the interval is the manifest's	
BlindLimitManifestTests	test_required_with_no_default	Required with no default.	
BlindLimitManifestTests	test_bounded_both_ways	Bounded both ways.	
BlindLimitManifestTests	test_at_least_three_intervals	At least three intervals.	
BlindLimitManifestTests	test_version_1_manifest_gets_a_migration_hint	Version 1 manifest gets a migration hint.	
GitWatchUnsettledTests	test_moving_refs_are_unsettled	Moving refs are unsettled.	

Class	Test	What it pins	Cites
GitWatchUnsettledTests	test_a_changing_worktree_is_unsettled	A changing worktree is unsettled.	
GitWatchUnsettledTests	test_output_over_the_cap_fails_the_cycle	Before r3.5 this was accepted as "dirty: null" every cycle, so it never reached the limit.	
GitWatchUnsettledTests	test_a_still_repository_is_observed	A still repository is observed.	
DirWatchCapacityTests	test_a_folder_over_the_cap_fails_the_cycle	A folder over the cap fails the cycle.	HF-33
DirWatchCapacityTests	test_adding_a_file_over_the_cap_never_reports_a_removal	Adding a file over the cap never reports a removal.	HF-33
DirWatchCapacityTests	test_at_the_cap_the_inventory_is_complete	At the cap the inventory is complete.	HF-33
NoRestartTests	test_fail_closed_exits_are_never_restarted	Fail closed exits are never restarted.	
NoRestartTests	test_a_stop_outlasts_one_watchdog_period	r3.9 (HF-31): a stop is honoured between cycles; a fixed 30 s SIGKILLED slow cycles.	HF-31
NoRestartTests	test_git_watch_worst_case_sense_fits_half_the_watchdog	r3.9 (HF-30): six Git calls per worktree cycle since r3.5, each up to step_timeout.	HF-30
NoRestartTests	test_uncertain_commit_stays_restartable	Uncertain commit stays restartable.	

tests/test_canonical.py (9 tests)

Canonical JSON: golden vector, refusals, determinism, strict parsing.

Class	Test	What it pins	Cites
CanonicalTests	test_golden_vector	Golden vector.	
CanonicalTests	test_genesis_is_64_zeros	Genesis is 64 zeros.	
CanonicalTests	test_key_order_matches_rfc8785_not_code_point_order	Key order matches rfc8785 not code point order.	
CanonicalTests	test_key_order_nested	The reordering must apply at every depth, not only the top level.	
CanonicalTests	test_allowed_conversions	Allowed conversions.	
CanonicalTests	test_refusals	Refusals.	
CanonicalTests	test_lone_surrogate_refused	Lone surrogate refused.	
CanonicalTests	test_independent_of_hash_seed	Independent of hash seed.	
CanonicalTests	test_strict_loads	Strict loads.	

tests/test_guard.py (8 tests)

Guard: output-directory checks, inventory, path policy, and the audit hook in a child process.

Class	Test	What it pins	Cites
OutputDirTests	test_creates_0700_tree	Creates 0700 tree.	
OutputDirTests	test_refuses_loose_permissions	Refuses loose permissions.	
OutputDirTests	test_refuses_symlink	Refuses symlink.	
OutputDirTests	test_stray_temp_files_removed	Stray temp files removed.	
OutputDirTests	test_inventory_reports_foreign_files	Inventory reports foreign files.	
AuditHookTests	test_enforce_mode_blocks_every_forbidden_operation	Enforce mode blocks every forbidden operation.	
AuditHookTests	test_record_mode_reports_without_blocking	Record mode reports without blocking.	
AuditHookTests	test_policy_readable	Policy readable.	

tests/test_handoff.py (32 tests)

The handoff layer: the published contract, the manifest JSON Schema, candidate envelopes,

Class	Test	What it pins	Cites
PublishedContractTests	test_committed_contract_matches_the_running_rules	Committed contract matches the running rules.	
PublishedContractTests	test_committed_schema_matches_the_running_validator	Committed schema matches the running validator.	
PublishedContractTests	test_contract_version_is_pinned_to_its_sha	Changing any enforced rule changes contract_sha256. That must come with a new CONTRACT_VERSION and a new line in contract/versions.json, never a silent edit.	
PublishedContractTests	test_contract_is_identical_on_every_installed_python	Contract is identical on every installed python.	
PublishedContractTests	test_contract_names_every_ctx_method_and_battery_check	Contract names every ctx method and battery check.	
PublishedContractTests	test_describe_cli	Describe cli.	
SchemaAgreementTests	test_schema_is_valid_draft_2020_12	Schema is valid draft 2020 12.	
SchemaAgreementTests	test_every_shipped_manifest_passes_both	Every shipped manifest passes both.	
SchemaAgreementTests	test_mutations_are_refused_by_both	Mutations are refused by both.	
SchemaAgreementTests	test_cross_field_rules_are_manifest_py_only_and_declared	Cross field rules are manifest py only and declared.	
EnvelopeTests	test_scaffold_envelope_is_exact	Scaffold envelope is exact.	
EnvelopeTests	test_a_candidate_cannot_certify_itself	A candidate cannot certify itself.	
EnvelopeTests	test_files_changed_after_sealing	Files changed after sealing.	
EnvelopeTests	test_other_major_version_is_incompatible	Other major version is incompatible.	
EnvelopeTests	test_same_version_other_rules_is_a_mismatch	Same version other rules is a mismatch.	
EnvelopeTests	test_same_major_other_minor_is_a_warning	Same major other minor is a warning.	
EnvelopeTests	test_bad_producer_kind	Bad producer kind.	
EnvelopeTests	test_envelope_cli_round_trip	Envelope cli round trip.	
ValidateJsonTests	test_structured_diagnostics	Structured diagnostics.	
ValidateJsonTests	test_manifest_diagnostics_carry_the_field	Manifest diagnostics carry the field.	
ValidateJsonTests	test_text_mode_is_unchanged_for_people	Text mode is unchanged for people.	
PrecheckTests	test_every_reference_daemon_prechecks_ok	Every reference daemon prechecks ok.	
PrecheckTests	test_impure_daemon_fails_fast_and_skips_the_run	Impure daemon fails fast and skips the run.	
PrecheckTests	test_policy_violation_is_caught_by_the_short_run	Policy violation is caught by the short run.	
PrecheckTests	test_precheck_never_says_pass	Precheck never says pass.	
PrecheckTests	test_an_absolute_output_dir_prechecks_like_the_battery	r4.7 (HF-37): the battery and precheck both rewrite an absolute output_dir into their workspace, but precheck bound the original manifest: "no ledger", and the rewritten folder reported as a write outside the output ...	HF-37
ScaffoldTests	test_refuses_a_non_empty_folder_and_bad_names	Refuses a non empty folder and bad names.	
ScaffoldTests	test_scaffold_passes_validate_and_user_unit_variant	Scaffold passes validate and user unit variant.	

Class	Test	What it pins	Cites
SymlinkStatTests	test_planted_symlink_is_a_link_not_a_violation	Planted symlink is a link not a violation.	HF-17
SymlinkStatTests	test_following_reads_still_judge_the_target	Following reads still judge the target.	HF-17
BatteryCatchesAlwaysFailingDaemonsTests	test_a_daemon_that_errors_every_cycle_fails_db04	r2 passed the raiser fixture: its sense() fails every cycle in the battery's workspace (no ~/data/mode.txt), yet the process exits 0 with a valid chain.	DB-04, HF-16
BatteryCatchesAlwaysFailingDaemonsTests	test_a_daemon_that_never_observes_fails_db20	r4.5 (HF-35): every cycle unsettled means no DAEMON_ERROR and no event; with its digest disabled, r4.4's battery reported RESULT: PASS for it.	DB-04, DB-20, HF-35

tests/test_hardening.py (40 tests)

Regression tests for the r3 hardening: every escape or crash found in the r2 review has a

Class	Test	What it pins	Cites
ManifestHardeningTests	test_interpreter_families_and_command_runners_are_refused	Interpreter families and command runners are refused.	HF-11
ManifestHardeningTests	test_harmless_commands_still_allowed	Harmless commands still allowed.	
ManifestHardeningTests	test_dot_segments_and_trailing_slashes_refused	Dot segments and trailing slashes refused.	HF-14
ManifestHardeningTests	test_output_dir_must_not_contain_a_protected_tree	r2 accepted "~": Landlock and ReadWritePaths= would then grant write access to the whole home directory, ~/.ssh and ~/spark-core included.	HF-12
ManifestHardeningTests	test_output_dir_inside_the_manifests_own_deny_is_refused	Output dir inside the manifests own deny is refused.	
ManifestHardeningTests	test_trailing_newlines_are_refused	r2 used re.match with "\$", which also matches before a final newline: a name of "x\n" split the generated unit's Description= line in two.	HF-13
ManifestHardeningTests	test_to_dict_is_bounded_and_strict	To dict is bounded and strict.	
GitHardeningTests	test_refusals	Refusals.	
GitHardeningTests	test_allowed	Allowed.	
GitHardeningTests	test_diff_subcommands_get_no_ext_diff_and_no_textconv	Diff subcommands get no ext diff and no textconv.	HF-08, HF-19, PD-25
GitHistoryIntegrityTests	test_repository_gpg_program_never_runs	Repository gpg program never runs.	
GitHistoryIntegrityTests	test_replace_refs_do_not_forge_history	Replace refs do not forge history.	
GitHistoryIntegrityTests	test_grafts_do_not_rewrite_ancestry	Grafts do not rewrite ancestry.	
GitInjectionRuntimeTests	test_alias_injection_through_ctx_git_is_refused	Alias injection through ctx git is refused.	
GitInjectionRuntimeTests	test_git_through_ctx_run_is_refused	Git through ctx run is refused.	HF-02
GitInjectionRuntimeTests	test_read_only_git_still_works	Read only git still works.	
PurityHardeningTests	test_frame_escape_to_real_builtins	Frame escape to real builtins.	HF-04
PurityHardeningTests	test_traceback_frame_escape	Traceback frame escape.	HF-04
PurityHardeningTests	test_private_attribute	Private attribute.	HF-03
PurityHardeningTests	test_aliasing_a_forbidden_builtin	Aliasing a forbidden builtin.	HF-05
PurityHardeningTests	test_dynamic_attribute_helpers	Dynamic attribute helpers.	HF-06
PurityHardeningTests	test_format_string_reaching_a_dunder	Format string reaching a dunder.	HF-06
PurityHardeningTests	test_exit_and_quit	Exit and quit.	
PurityHardeningTests	test_import_time_calls	Import time calls.	

Class	Test	What it pins	Cites
PurityHardeningTests	test_entry_points_cannot_be_reassigned_or_redefined	Entry points cannot be reassigned or redefined.	
PurityHardeningTests	test_allowed_idioms_still_pass	Allowed idioms still pass.	
PurityHardeningTests	test_every_shipped_daemon_is_still_pure	Every shipped daemon is still pure.	
RuntimeHardeningTests	test_system_exit_in_sense_is_recorded_not_a_silent_exit	r2: raise SystemExit(0) ended the process with exit 0 and no DAEMON_STOP, so systemd's Restart=on-failure would never have restarted it.	HF-09
RuntimeHardeningTests	test_undeclared_command_request_fails_closed_even_when_swallowed	Undeclared command request fails closed even when swallowed.	HF-10
RuntimeHardeningTests	test_rejected_duplicate_launch_touches_nothing	r3.2 (WBS 3.1 §4.2): the second instance cleaned tmp/ before taking the lock and deleted the running instance's in-flight files (for git-watch, its index copy).	HF-25
RuntimeHardeningTests	test_torn_first_write_still_starts_the_ledger_with_daemon_start	r3.2 (WBS 3.0 r4's pre-baseline correction): a crash during the first-ever append left bytes but no complete record; the next start wrote LEDGER_TAIL_QUARANTINED as seq 1, a ledger the battery's own DB-04 then ...	DB-04, HF-26
RuntimeHardeningTests	test_policy_is_immutable	Policy is immutable.	HF-03
CpuAccountingTests	test_cpu_includes_commands_the_daemon_ran	Cpu includes commands the daemon ran.	HF-18
UnitHardeningTests	test_unit_allows_the_landlock_syscalls	Unit allows the landlock syscalls.	HF-07
UnitHardeningTests	test_resolve_syscall_filter	Resolve syscall filter.	HF-07
UnitHardeningTests	test_unsafe_paths_are_refused	Unsafe paths are refused.	HF-22
RenderLabelTests	test_label_with_trailing_newline_is_refused	Label with trailing newline is refused.	HF-13
RenderEfficiencyTests	test_binary_search_matches_the_linear_scan	Binary search matches the linear scan.	HF-21
CliRobustnessTests	test_verify_with_a_bad_manifest_reports_instead_of_crashing	Verify with a bad manifest reports instead of crashing.	
CliRobustnessTests	test_battery_with_invalid_json_reports_instead_of_crashing	Battery with invalid json reports instead of crashing.	

tests/test_landlock.py (13 tests)

Landlock (AP-01): ABI detection, the pure helpers, and real enforcement in forked

Class	Test	What it pins	Cites
AbiTests	test_abi_version_is_an_int	Abi version is an int.	
AbiTests	test_this_workspace_kernel_supports_landlock	Documents the environment this suite runs in; the DGX (aarch64, a different kernel) must be checked separately per ADJUDICATION-AP.md.	
PureHelperTests	test_fs_access_mask_grows_with_abi	Fs access mask grows with abi.	
PureHelperTests	test_fs_access_for_file_drops_directory_only_rights	Fs access for file drops directory only rights.	
PureHelperTests	test_fs_access_for_directory_keeps_every_requested_right	Fs access for directory keeps every requested right.	
PureHelperTests	test_gaps_finds_a_denied_path_inside_a_granted_read	Gaps finds a denied path inside a granted read.	
PureHelperTests	test_gaps_empty_when_reads_and_denies_do_not_overlap	Gaps empty when reads and denies do not overlap.	
SupervisorDomainTestModeTests	test_unavailable_landlock_is_non_fatal_in_test_mode	Unavailable landlock is non fatal in test mode.	
SupervisorDomainTestModeTests	test_unavailable_landlock_refuses_outside_test_mode	Unavailable landlock refuses outside test mode.	
EnforcementTests	test_supervisor_domain_blocks_denied_reads_and_outside_writes	Supervisor domain blocks denied reads and outside writes.	

Class	Test	What it pins	Cites
EnforcementTests	test_nested_child_domain_can_only_narrow_never_widen	Nested child domain can only narrow never widen.	AP-04
EnforcementTests	test_nothing_the_daemon_can_write_or_merely_reads_can_be_executed	r4.6 (HF-36): the domain granted EXECUTE on the output directory and on every declared read path, so if the in-process layers were bypassed, a binary written into the output directory, or dropped by someone else into ...	HF-36
EnforcementTests	test_output_directory_gets_every_right_the_abi_handles	Output directory gets every right the abi handles.	

tests/test_layers.py (4 tests)

Framework / services boundary (r3.4, PD-53; a Python reading of the Asterinas framekernel

Class	Test	What it pins	Cites
LayerBoundaryTests	test_service_modules_import_nothing_that_reaches_the_os	Service modules import nothing that reaches the os.	
LayerBoundaryTests	test_service_modules_never_call_open	Service modules never call open.	
LayerBoundaryTests	test_ctypes_only_where_declared	Ctypes only where declared.	
LayerBoundaryTests	test_exactly_one_truncate_site	r4's audit rule: truncation destroys bytes, so it happens in one reviewed place (ledger.recover, after the torn tail is copied to quarantine).	

tests/test_ledger.py (16 tests)

Ledger: append and verify, every corruption category, torn tails, uncertain commits.

Class	Test	What it pins	Cites
LedgerTests	test_append_and_verify	Append and verify.	
LedgerTests	test_flipped_byte	Flipped byte.	
LedgerTests	test_swapped_lines	Swapped lines.	
LedgerTests	test_missing_line	Missing line.	
LedgerTests	test_non_canonical_line	Non canonical line.	
LedgerTests	test_foreign_daemon	Foreign daemon.	
LedgerTests	test_newer_schema_fails_explicitly	Newer schema fails explicitly.	
LedgerTests	test_overlong_line	Overlong line.	
LedgerTests	test_corrupt_last_complete_line_is_not_a_torn_tail	Corrupt last complete line is not a torn tail.	
LedgerTests	test_torn_tail_garbage_is_quarantined	Torn tail garbage is quarantined.	
LedgerTests	test_complete_record_without_newline_is_quarantined	Complete record without newline is quarantined.	
LedgerTests	test_two_torn_tails_never_overwrite	Two torn tails never overwrite.	
LedgerTests	test_prepare_refuses_before_writing	Prepare refuses before writing.	
LedgerTests	test_fsync_failure_is_uncertain	Fsync failure is uncertain.	
LedgerTests	test_short_write_is_uncertain	Short write is uncertain.	
LedgerTests	test_streaming_pass_memory_does_not_depend_on_length	Streaming pass memory does not depend on length.	

tests/test_manifest.py (15 tests)

Manifest: the example validates; every refusal category is enforced.

Class	Test	What it pins	Cites
ManifestTests	test_example_is_valid	Example is valid.	
ManifestTests	test_hash_ignores_key_order_and_whitespace	Hash ignores key order and whitespace.	
ManifestTests	test_unknown_keys_refused_at_every_level	Unknown keys refused at every level.	
ManifestTests	test_missing_key	Missing key.	
ManifestTests	test_acting_daemons_refused	Acting daemons refused.	
ManifestTests	test_network_refused	Network refused.	

Class	Test	What it pins	Cites
ManifestTests	test_forbidden_commands	Forbidden commands.	
ManifestTests	test_reads_inside_denied_paths	Reads inside denied paths.	
ManifestTests	test_output_dir_placement	Output dir placement.	
ManifestTests	test_reserved_event_types	Reserved event types.	
ManifestTests	test_step_timeout_bounded_by_watchdog	Step timeout bounded by watchdog.	
ManifestTests	test_system_unit_needs_non_root_user	System unit needs non root user.	
ManifestTests	test_paths_limited_to_safe_characters	Paths limited to safe characters.	
ManifestTests	test_purpose_cannot_carry_systemd_specifiers	Purpose cannot carry systemd specifiers.	
ManifestTests	test_floats_refused	Floats refused.	

tests/test_misc.py (14 tests)

Unit generator, purity check, battery and source hygiene.

Class	Test	What it pins	Cites
UnitgenTests	test_example_unit_carries_every_required_directive	Example unit carries every required directive.	
UnitgenTests	test_lint_detects_a_removed_directive	Lint detects a removed directive.	
UnitgenTests	test_user_units_carry_a_warning	User units carry a warning.	
UnitgenTests	test_a_required_path_becomes_a_start_condition	r4.8 (from the first pilot): with a locked drive, a daemon whose output is on it must be skipped by systemd, not stopped fail-closed (78, never restarted).	
UnitgenTests	test_part_of_binds_the_unit_to_another	r4.8 (from the first pilot): an open ledger keeps a drive busy, so a daemon writing to a drive another unit owns must stop before it and start with it, never at boot.	
UnitgenTests	test_install_plan_is_text_only	Install plan is text only.	
PurityTests	test_good_module	Good module.	
PurityTests	test_example_is_pure	Example is pure.	
PurityTests	test_each_forbidden_construct	Each forbidden construct.	
PurityTests	test_required_functions_and_arity	Required functions and arity.	
BatteryTests	test_example_passes_or_is_incomplete_never_fails	Example passes or is incomplete never fails.	DB-18, DB-19, DB-20, HF-35, PD-76
BatteryTests	test_impure_daemon_fails	Impure daemon fails.	DB-02, DB-03
RuntimeStructureTests	test_gc_collect_runs_once_per_cycle_not_inside_the_sleep_loop	Structural guard for the multi-daemon GC-forcing addition (r2): gc.collect() must be called exactly once per cycle of the outer loop, and not inside the inner sleep loop (which would call it many times per cycle, ...	
SourceHygieneTests	test_no_raw_invisible_characters_in_source	Trojan-source guard: bidi, zero-width and control characters appear only as escapes.	

tests/test_proc.py (6 tests)

Hardened subprocess: allowlist, scrubbed environment, bounds, timeout, and the F4 index-copy fix.

Class	Test	What it pins	Cites
ProcTests	test_only_allowlisted_commands_run	Only allowlisted commands run.	
ProcTests	test_output_is_bounded	Output is bounded.	
ProcTests	test_timeout_kills_the_group	Timeout kills the group.	
ProcTests	test_git_ignores_user_configuration	Script-board F5: color.ui=always and diff.external in the user's config must not leak.	F5

Class	Test	What it pins	Cites
ProcTests	test_inherited_git_selection_variables_cannot_redirect_git	r4.0 (hardening review, item 2): GIT_DIR, GIT_WORK_TREE and GIT_OBJECT_DIRECTORY in the daemon's own environment must not redirect a call to a decoy repository. Every child starts from SAFE_ENV (an allowlist), so ...	
ProcTests	test_index_copy_leaves_the_real_index_untouched	Script-board F4: porcelain git diff rewrites .git/index; the index copy does not.	F4

tests/test_render.py (10 tests)

Rendering: hostile values stay inside their blocks; structure never changes; bounds hold.

Class	Test	What it pins	Cites
RenderTests	test_hostile_corpus_is_inert	Hostile corpus is inert.	
RenderTests	test_structure_invariance_property	Structure invariance property.	
RenderTests	test_escapes_are_visible_and_unambiguous	Escapes are visible and unambiguous.	
RenderTests	test_multiline_values_are_prefixed	Multiline values are prefixed.	
RenderTests	test_fence_longer_than_any_backtick_run	Fence longer than any backtick run.	
RenderTests	test_labels_are_validated	Labels are validated.	
RenderTests	test_values_are_type_checked	Values are type checked.	
RenderTests	test_size_bound_truncates_with_marker	Size bound truncates with marker.	
RenderTests	test_stamp_is_validated	Stamp is validated.	
RenderTests	test_deterministic	Deterministic.	

tests/test_runtime.py (18 tests)

Runtime end to end: lifecycle, errors, fail-closed policy, the watchdog, refusals.

Class	Test	What it pins	Cites
RuntimeTests	test_lifecycle_and_digest	Lifecycle and digest.	
RuntimeTests	test_hostile_value_is_contained_in_the_digest	Hostile value is contained in the digest.	
RuntimeTests	test_errors_are_bounded_and_streaks_suppressed	Errors are bounded and streaks suppressed.	
RuntimeTests	test_error_streak_clears_on_success	Error streak clears on success.	
RuntimeTests	test_policy_violation_fails_closed_even_when_swallowed	Policy violation fails closed even when swallowed.	
RuntimeTests	test_undeclared_event_type_is_refused	Undeclared event type is refused.	
RuntimeTests	test_float_payload_is_refused	Float payload is refused.	
RuntimeTests	test_test_mode_refused_under_systemd	Test mode refused under systemd.	
RuntimeTests	test_impure_code_never_starts	Impure code never starts.	
RuntimeTests	test_unsafe_output_directory_is_refused	Unsafe output directory is refused.	
RuntimeTests	test_foreign_files_are_reported_not_touched	Foreign files are reported not touched.	
RuntimeTests	test_corrupt_ledger_refuses_and_changes_nothing	Corrupt ledger refuses and changes nothing.	
RuntimeTests	test_watchdog_pings_stop_while_an_observation_hangs	No background pinger: a blocked sense() starves the watchdog, so systemd would restart it.	
RuntimeTests	test_jitter_bound_is_zero_in_test_mode	Jitter bound is zero in test mode.	
RuntimeTests	test_jitter_bound_is_a_fraction_of_the_interval	Jitter bound is a fraction of the interval.	
RuntimeTests	test_jitter_bound_is_capped_for_long_intervals	86400s (the manifest maximum) times 5% would be 4320s; it must not drift by hours.	
RuntimeTests	test_jitter_disabled_in_test_mode_and_recorded_for_a_real_run	Jitter disabled in test mode and recorded for a real run.	
RuntimeTests	test_example_daemon_runs	Example daemon runs.	